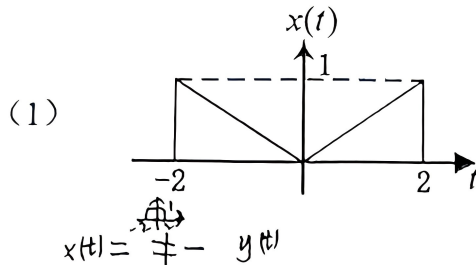


信号与系统第二次测试 (2026 年)

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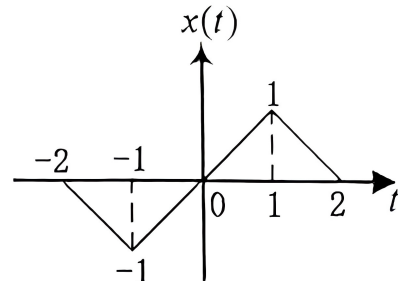
学号: 3240101517 30

1. 求如下信号的傅里叶变换



$y(t) \xrightarrow{F} 2\text{Sa}^2(\omega)$
 $\xrightarrow{F} 4\text{Sa}(\omega)$
 $x(j\omega) = \frac{1}{4\text{Sa}(\omega)} - 2\text{Sa}^2(\omega)$

(2)

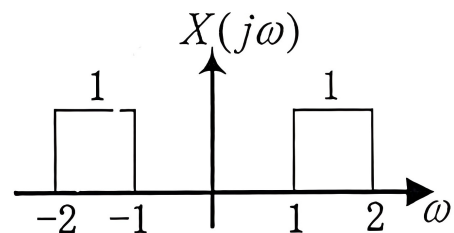
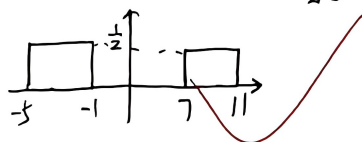


$y(t) \xrightarrow{F} \text{Sa}^2(\frac{\omega}{2})$
 $y(t-1) \xrightarrow{F} \text{Sa}^2(\frac{\omega}{2}) e^{-j\omega}$
 $-y(t+1) \xrightarrow{F} -\text{Sa}^2(\frac{\omega}{2}) e^{j\omega}$
 $x(t) \xrightarrow{F} -2j \text{Sa}^2(\frac{\omega}{2}) \sin \omega$

2. 求如下信号的傅里叶反变换

$(1) X(j\omega) = \frac{2\sin(2\omega)\cos(6\omega)}{\omega} e^{-j3\omega}$
 $X(j\omega) = \frac{2\sin 2\omega}{\omega} (e^{j3\omega} + e^{-j9\omega})$

$y(t) \xrightarrow{F} 2\text{Sa}(2\omega) = \frac{\sin 2\omega}{\omega}$
 $x(t) = y(t+3) + y(t-9) = \frac{1}{2} [u(t-7) - u(t-11)] + \frac{1}{2} [u(t+5) - u(t+1)]$



$x(t) = \frac{\sin 2t - \sin t}{\pi t}$
 $X(j\omega) = \frac{1}{\pi} [\text{rect}(\frac{\omega+1.5}{2}) - \text{rect}(\frac{\omega-1.5}{2})]$

3. 已知微分方程

$$\frac{d^2 y(t)}{dt^2} + 5 \frac{dy(t)}{dt} + 6y(t) = \frac{dx(t)}{dt} + x(t)$$

试求当 $x(t) = e^{-4t}u(t)$ 时, 系统的零状态响应 $y(t)$ 。

$$H(j\omega) = \frac{j\omega + 1}{(j\omega)^2 + 5j\omega + 6}$$

$$X(j\omega) = \frac{1}{j\omega + 4}$$

$$Y(j\omega) = \frac{j\omega + 1}{(j\omega + 4)(j\omega + 2)(j\omega + 3)}$$

$$= -\frac{3}{2} \frac{1}{j\omega + 4} - \frac{1}{2} \frac{1}{(j\omega + 2)} + 2 \frac{1}{j\omega + 3}$$

$$y(t) = \cancel{2e^{-jt}} \left(2e^{-3t} - \frac{1}{2}e^{-2t} - \frac{3}{2}e^{-4t} \right) u(t)$$